MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 7

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

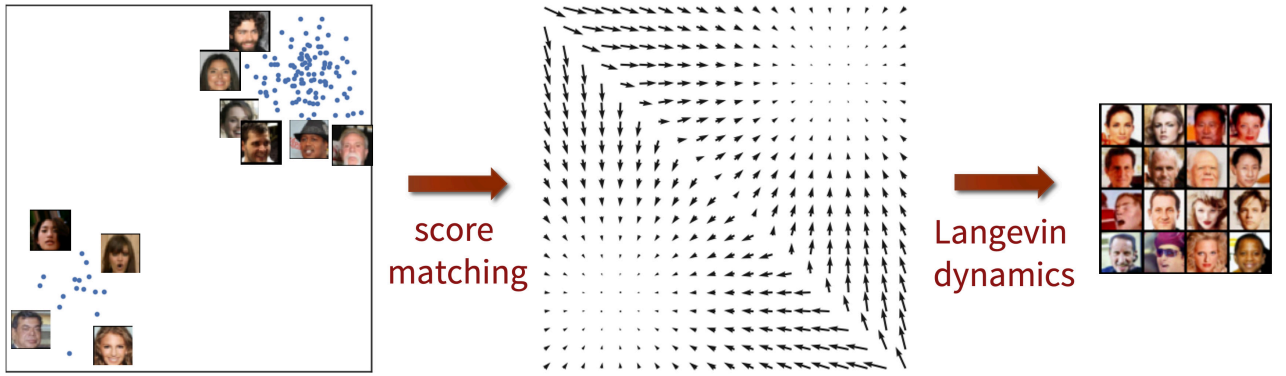
For an EBM $p_{\boldsymbol{\theta}}(\mathbf{x}) \propto e^{-E_{\boldsymbol{\theta}}(\mathbf{x})}$, its **score** is $\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) = -\nabla_{\mathbf{x}} E_{\boldsymbol{\theta}}(\mathbf{x})$.
We learn a vector network $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ directly, then sample with noise.

Fisher Divergence

$$D_F(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$



Recap of Previous Lecture

Let us perturb the original data with Gaussian noise $q(\mathbf{x}_\sigma|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$.

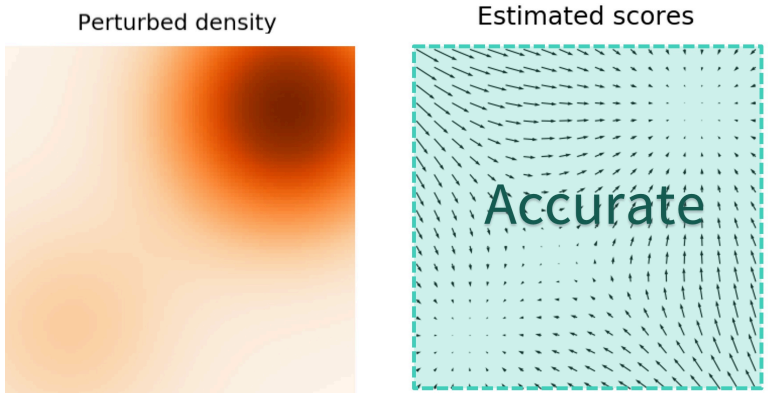
$$q(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}.$$

Assumption

The solution to

$$\frac{1}{2}\mathbb{E}_{q(\mathbf{x}_\sigma)}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

satisfies $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta},0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ if σ is sufficiently small.



Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021
Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Recap of Previous Lecture

Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\begin{aligned}&\mathbb{E}_{q(\mathbf{x}_\sigma)}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)\right\|_2^2 \\ &\quad= \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 + \text{const}\end{aligned}$$

Here,

$$\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x}) = -\frac{\mathbf{x}_\sigma - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}.$$

Tweedie's Formula

From Lecture 6, the MSE-optimal clean-data prediction is

$$\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] = \mathbf{x}_\sigma + \sigma^2\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma).$$

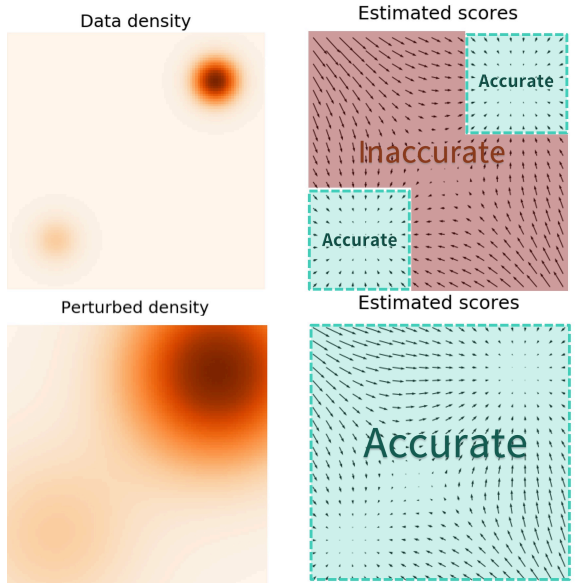
Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010
Efron B. Tweedie's Formula and Selection Bias, 2011

Recap of Previous Lecture

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathcal{N}(0,\mathbf{I})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}+\sigma\boldsymbol{\epsilon})+\frac{\boldsymbol{\epsilon}}{\sigma}\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2}\cdot\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta}\cdot\boldsymbol{\epsilon}_l$$

- For **small** σ, $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x})$ becomes inaccurate and Langevin dynamics fails to traverse modes
- For **large** σ, robustness in low-density regions is achieved, but the model learns a distribution that is overly corrupted



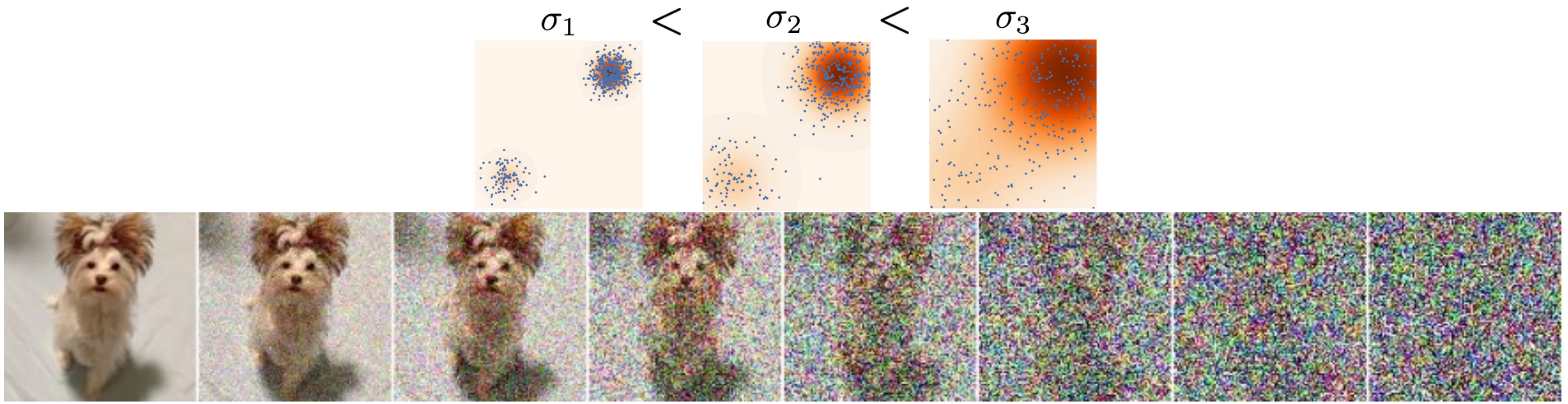
Recap of Previous Lecture

Noise-Conditioned Score Network (NCSN)

- Define a sequence of noise levels: $\sigma_1 < \sigma_2 < \cdots < \sigma_T$.
- Balance the score targets $-\boldsymbol{\epsilon}/\sigma_t$ using weights σ_t^2:

$$\sum_{t=1}^{T} \sigma_t^2 \cdot \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

- Sample using **annealed** Langevin dynamics (for $t = T, \ldots, 1$).



Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

Recap of Previous Lecture

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L} = \sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta}, \sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2$.
4. Update $\boldsymbol{\theta}$ via stochastic gradient descent.

Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via L steps of Langevin dynamics at each noise level σ_t:

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \mathbf{s}_{\boldsymbol{\theta}, \sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \boldsymbol{\epsilon}_t^l.$$

3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to σ_{t-1} if $t > 1$.

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019
Song Y. et al. Improved Techniques for Training Score-Based Generative Models, 2020

Outline

01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

05 Gaussian Diffusion Reparametrization

06 Denoising Diffusion Probabilistic Model (DDPM)

Outline

01 **Forward Gaussian Diffusion Process**

02 Reverse Gaussian Diffusion Process

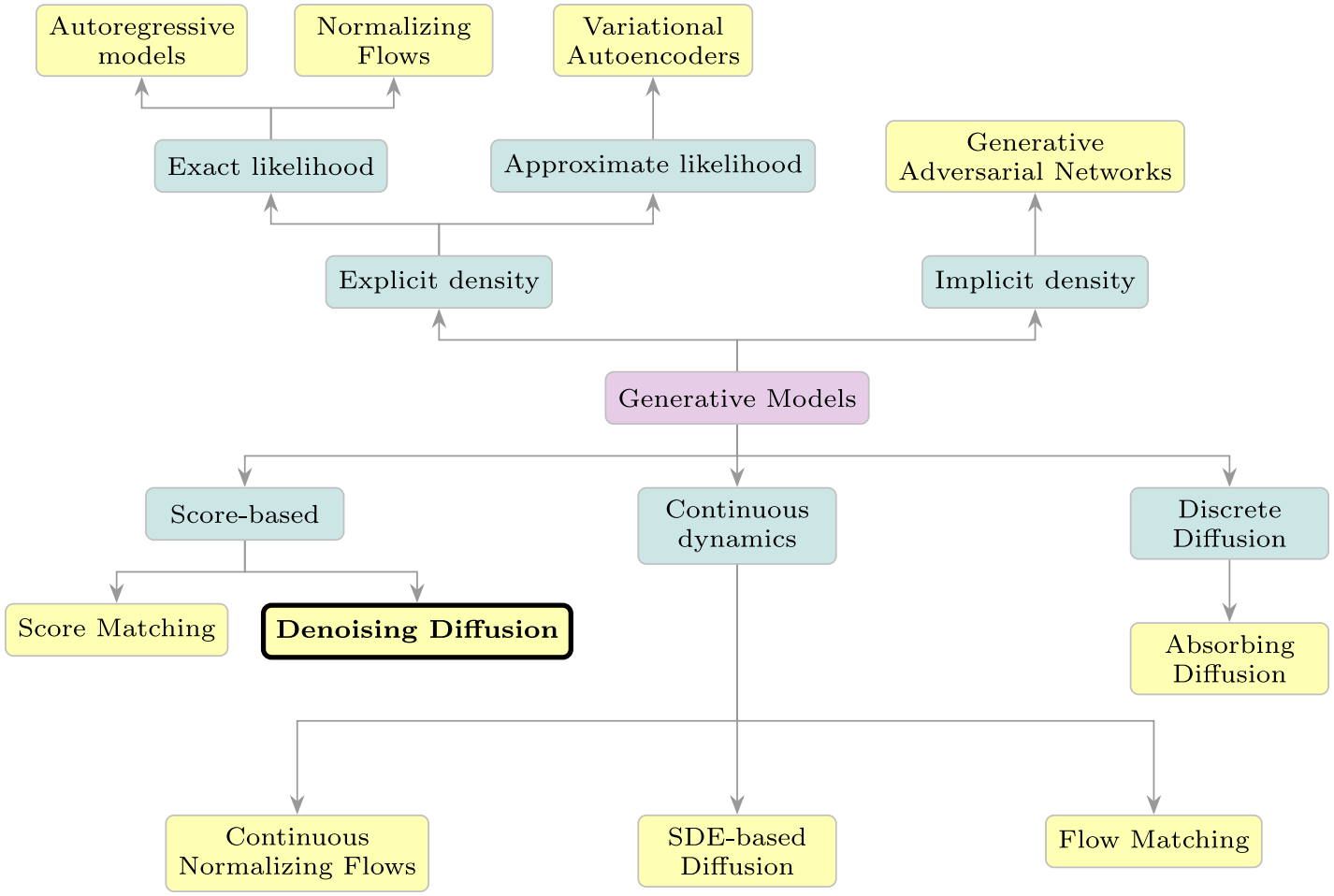
03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

05 Gaussian Diffusion Reparametrization

06 Denoising Diffusion Probabilistic Model (DDPM)

Generative Models Taxonomy



Forward Gaussian Diffusion Process

Let $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\beta_t \ll 1$. Define a Markov chain:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&\approx \left(1 - \frac{\beta_t}{2}\right)\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&= \mathbf{x}_{t-1} + \frac{\beta_t}{2}(-\mathbf{x}_{t-1}) + \sqrt{\beta_t}\boldsymbol{\epsilon}_t
\end{aligned}$$

From a Markov chain to Langevin dynamics: for small β_t, this Gaussian chain approximates LD targeting $\mathcal{N}(0, \mathbf{I})$, with step size $\eta = \beta_t$ and Gaussian score $-\mathbf{x}_{t-1}$.

Forward Diffusion: Sample at Any Timestep

Fix a schedule $0 < \beta_t < 1$. The noises $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$ are independent across t and of $\mathbf{x}_0$.

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \qquad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}).$$

Exact Conditional Distribution

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. Then

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}).$$

Sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$, without simulating the intermediate steps:

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t}\left(\sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_{t-1}}\boldsymbol{\epsilon}_{t-1}\right) + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + (\sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{\epsilon}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t) \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}'_t \\
&= \ldots = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).
\end{aligned}$$

Noise Variance

Independent noises: variances add per coordinate.

$$\begin{aligned}
&\alpha_t(1-\alpha_{t-1}) + (1-\alpha_t) \\
&\quad = 1 - \alpha_t\alpha_{t-1}. \\
&\boldsymbol{\epsilon}'_t \sim \mathcal{N}(0, \mathbf{I}).
\end{aligned}$$

When Does Only Gaussian Noise Remain?

The closed-form conditional already tells us how the signal disappears:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad \boldsymbol{\epsilon} \perp \mathbf{x}_0.$$

The Schedule Must Remove the Signal

For $0 < \beta_t < 1$, the required condition is

$$\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s) \longrightarrow 0 \quad \Longleftrightarrow \quad \sum_{t=1}^{\infty}\beta_t = \infty.$$

Many steps alone are not enough: $\beta_t = 2^{-t}$ leaves $\bar{\alpha}_\infty > 0$.

$$\sqrt{\bar{\alpha}_t} \to 0, \qquad \sqrt{1-\bar{\alpha}_t} \to 1 \quad \Longrightarrow \quad \mathbf{x}_t \xrightarrow{d} \mathcal{N}(0, \mathbf{I}).$$

The limit no longer depends on the initial data distribution p_{data}.

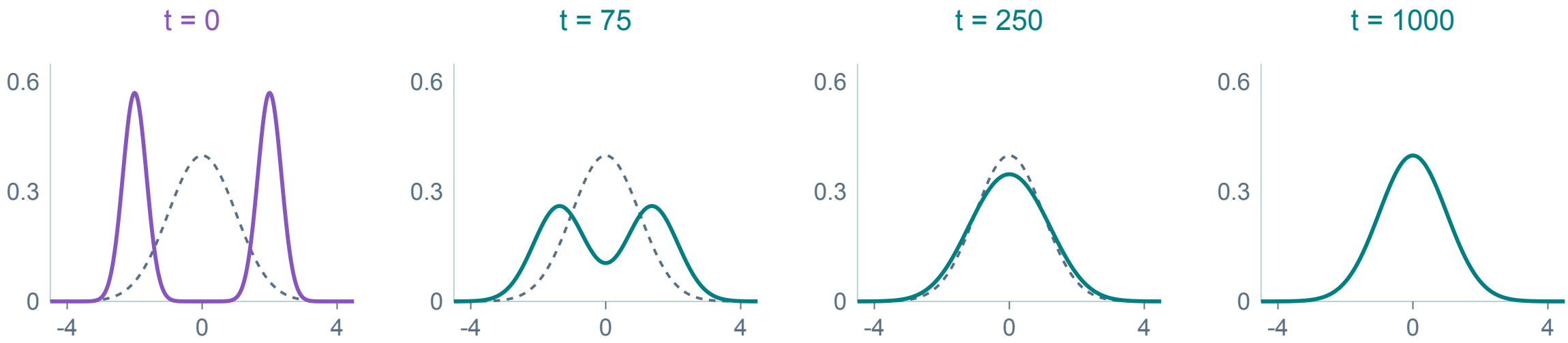
> **For finite T:** choose $\bar{\alpha}_T \ll 1$ with negligible residual signal $\sqrt{\bar{\alpha}_T}\mathbf{x}_0$.
> Then $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$: the terminal distribution is approximately standard Gaussian.

Forward Diffusion: From Data to Noise

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right);$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$p_{\text{data}}(x_0) = \tfrac{1}{2}\mathcal{N}(-2, 0.35^2) + \tfrac{1}{2}\mathcal{N}(2, 0.35^2), \qquad \beta_t = 0.01$$



---- Standard Gaussian reference $\mathcal{N}(0, 1)$.

The plots show marginal densities; the kernels above are conditional.

Denoising Score Matching

Let $\mathbf{x}_0 \sim p_{\text{data}}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$, with $\boldsymbol{\epsilon}$ independent of $\mathbf{x}_0$.

NCSN

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \mathbf{x}_0}{\sigma_t^2}$$

$$q(\mathbf{x}_1) \approx p_{\text{data}}, \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

Gaussian Diffusion

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1 - \bar{\alpha}_t}$$

$$q(\mathbf{x}_1) \approx p_{\text{data}}, \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

Theorem (Denoising Score Matching)

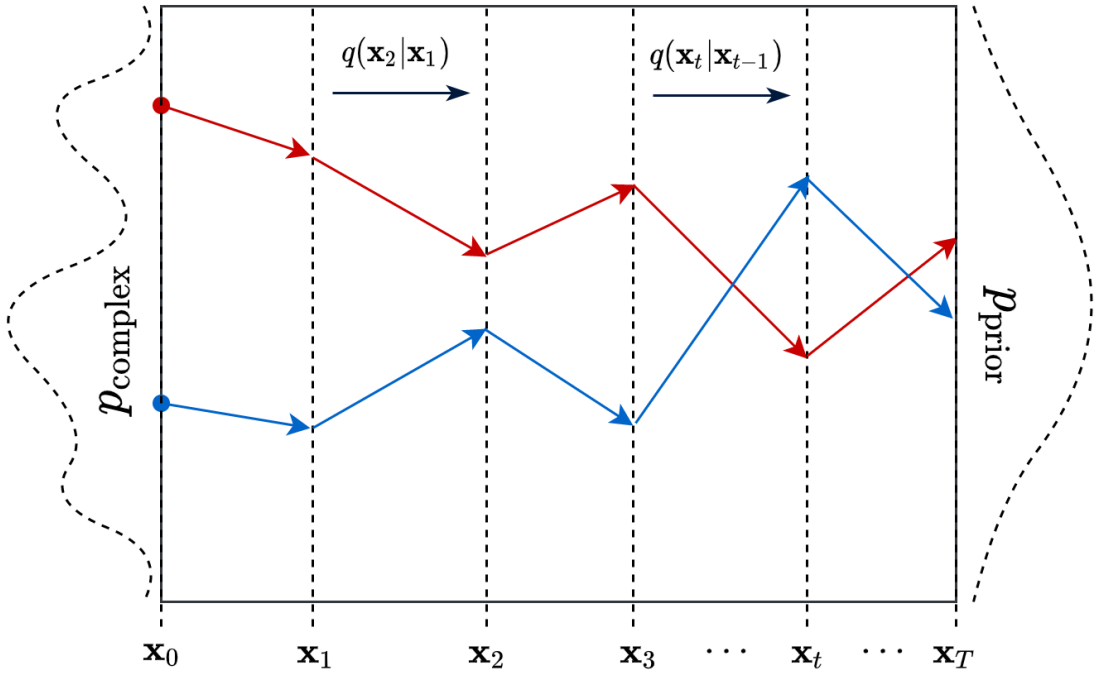
$$\mathbb{E}_{q(\mathbf{x}_t)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) \right\|_2^2 + \text{const}$$

Note: Annealed Langevin dynamics applies to diffusion, too.
NCSN with annealed Langevin dynamics is an early score-based generative framework (2019–2020).

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019
Sohl-Dickstein J. Deep Unsupervised Learning using Nonequilibrium Thermodynamics, 2015

Forward Gaussian Diffusion Process

Diffusion describes the migration of particles from regions of high density to those of low density.



1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t,$
   $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}),\ t \geq 1$
3. Choose $\bar{\alpha}_T \ll 1$ to suppress $\sqrt{\bar{\alpha}_T}\mathbf{x}_0$.
   Then $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$.

If this process can be reversed, we can sample from $p_{\text{data}}(\mathbf{x})$ by starting from noise $\mathcal{N}(0, \mathbf{I}) \approx q(\mathbf{x}_T)$.
Our goal now becomes inverting this diffusion.

Outline

01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

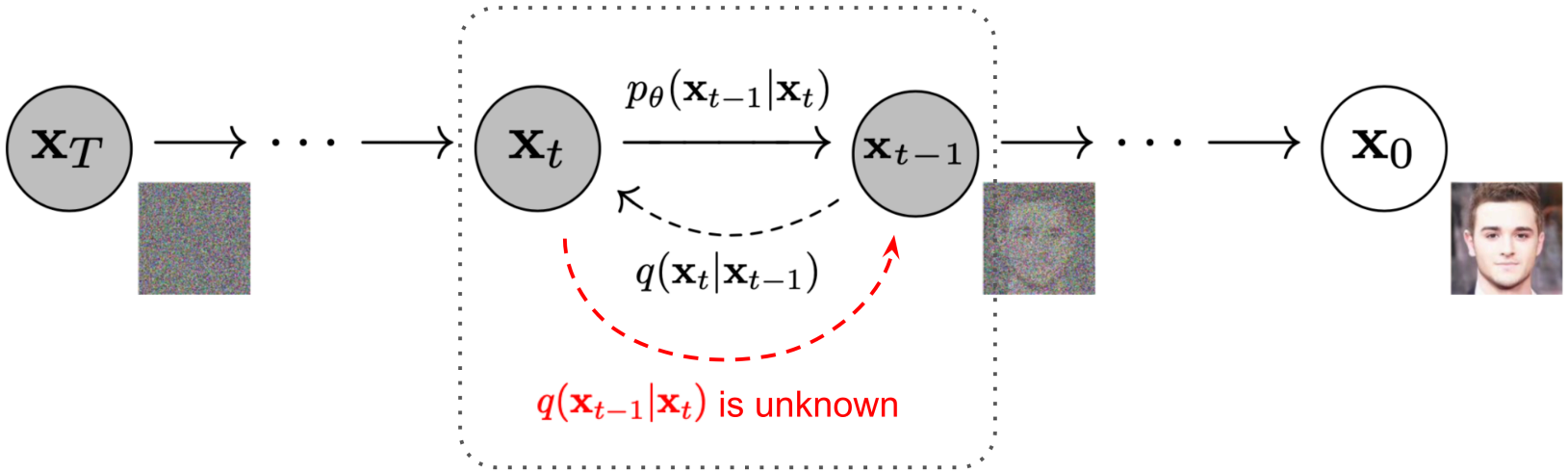
03 Gaussian Diffusion Model as VAE

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Reverse Gaussian Diffusion Process



Forward Process $$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right)$$

Reverse Process $$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

$q(\mathbf{x}_{t-1})$ and $q(\mathbf{x}_t)$ are intractable:

$$q(\mathbf{x}_t) = \int q(\mathbf{x}_t|\mathbf{x}_0)p_{\text{data}}(\mathbf{x}_0)d\mathbf{x}_0$$

Weng L. What are Diffusion Models?, blog post, 2021

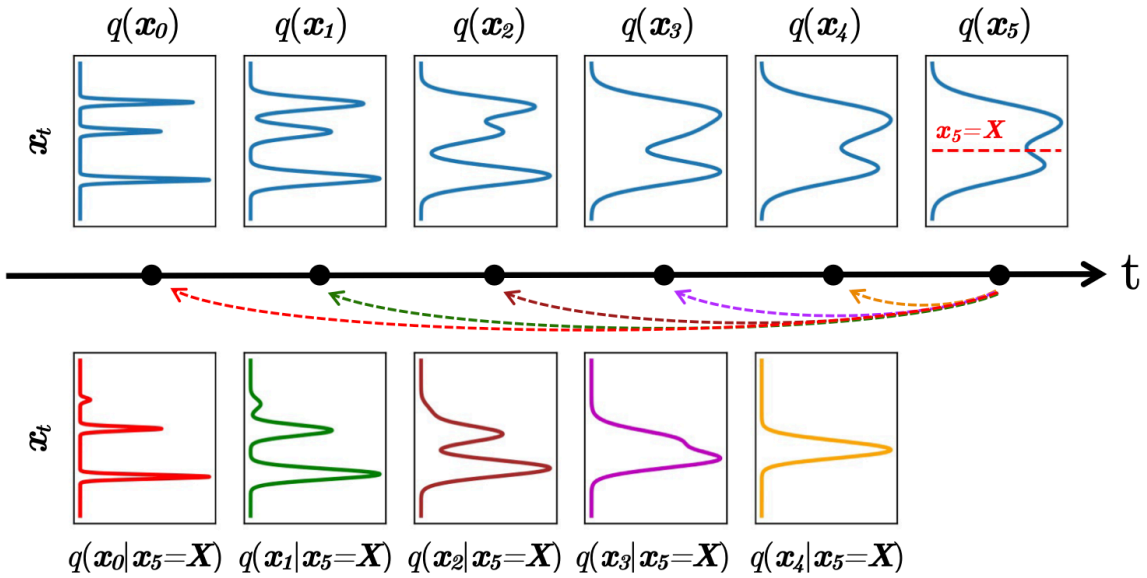
Reverse Gaussian Diffusion Process

Theorem (Feller, 1949)

If β_t is sufficiently small, the reverse conditional is approximately Gaussian:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)}.$$

The original DDPM used $T = 1000$ steps. Faster sampling methods will be discussed later.



Same Noisy Endpoint

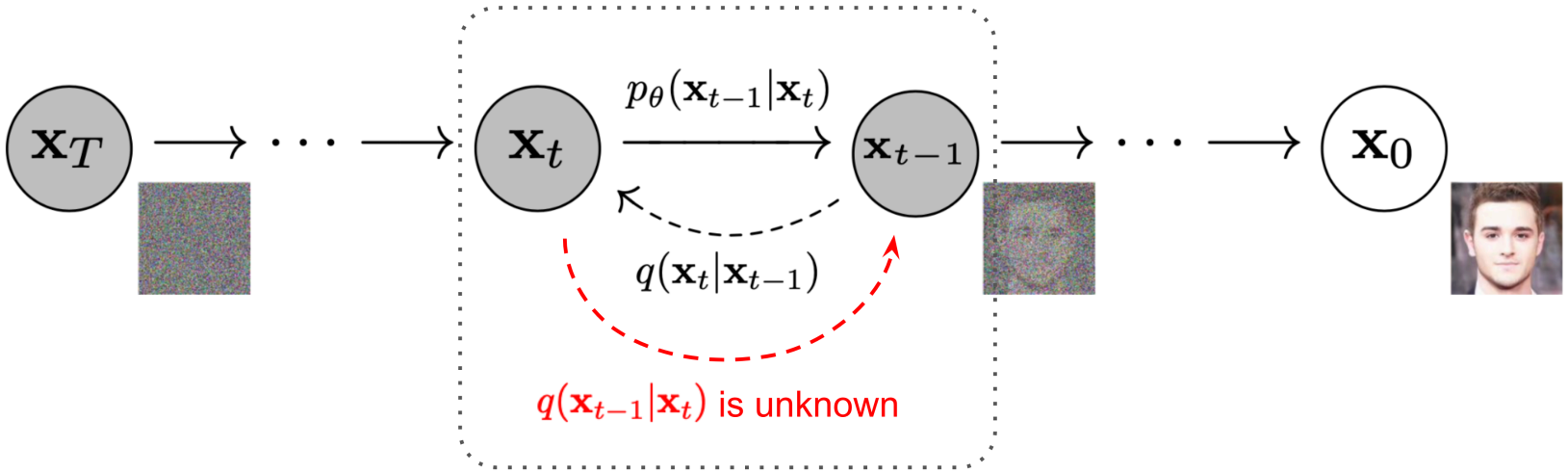
Top: marginals $q(x_s)$.
Bottom: posteriors for fixed $x_5 = X$.

$$q(x_s|x_5 = X) \propto q(x_5 = X|x_s)q(x_s).$$

- **Short reverse step:** nearly Gaussian.
- **Long reverse jump:** can be multimodal.

Feller W. On the theory of stochastic processes, with particular reference to applications, 1949
Ho J. et al. Denoising Diffusion Probabilistic Models, 2020
Xiao Z., Kreis K., Vahdat A. Tackling the generative learning trilemma with denoising diffusion GANs, 2021

Reverse Gaussian Diffusion Process (Ancestral Sampling)

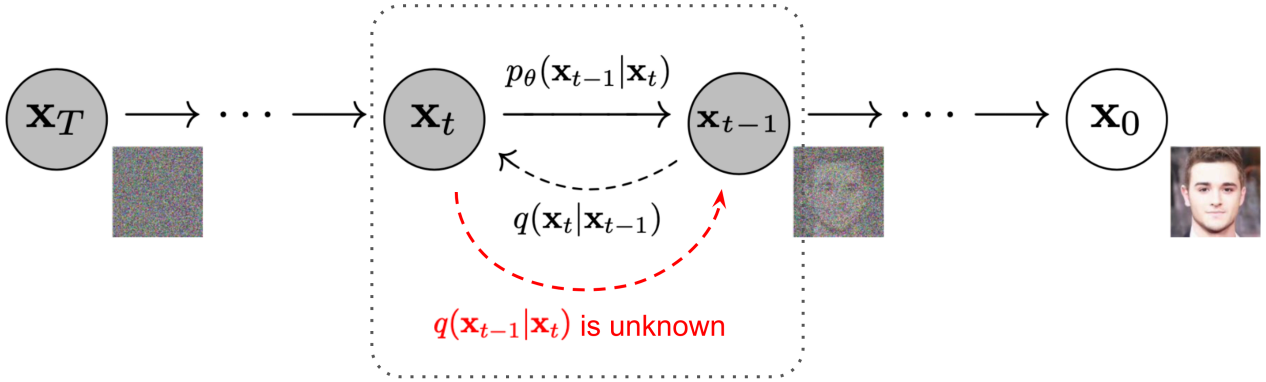


Forward Process $$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right)$$

Reverse Process $$q(\mathbf{x}_{t-1}|\mathbf{x}_t) \approx p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

If β_t is sufficiently small, $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is approximately Gaussian.

Reverse Gaussian Diffusion Process (Ancestral Sampling)



Forward Process

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right)$$

1. $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$
2. $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t$
3. $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$

Reverse Process

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

1. $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$
2. $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \boldsymbol{\sigma}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\boldsymbol{\epsilon}_t$
3. $\mathbf{x}_0 \sim p_{\boldsymbol{\theta}}(\mathbf{x}_0) \approx p_{\text{data}}(\mathbf{x}_0)$

Note: $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$. The forward process has no trainable parameters.

Conditioned Reverse Distribution

Reverse Kernel (Intractable)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}\big(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\big), \quad \beta_t \ll 1.$$

Conditioned Reverse Kernel (Tractable)

During training, $\mathbf{x}_0$ is known: substitute Gaussians and complete the square in $\mathbf{x}_{t-1}$.

$$\begin{aligned} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\ &= \frac{\mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})\,\mathcal{N}(\mathbf{x}_{t-1}; \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0, (1-\bar{\alpha}_{t-1})\mathbf{I})}{\mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})} \\ &= \mathcal{N}\big(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t\mathbf{I}\big). \end{aligned}$$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t} \cdot \mathbf{x}_0$$

$$\tilde{\beta}_t = \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} = \text{const}$$

What Does Knowing the Clean Data Change?

Same observation $x_t = 0$: unknown clean data versus each known clean value.

Two possible clean values

$\mathbb{P}(x_0 = -2) = \mathbb{P}(x_0 = 2) = \frac{1}{2}$

$x_0 = -2$

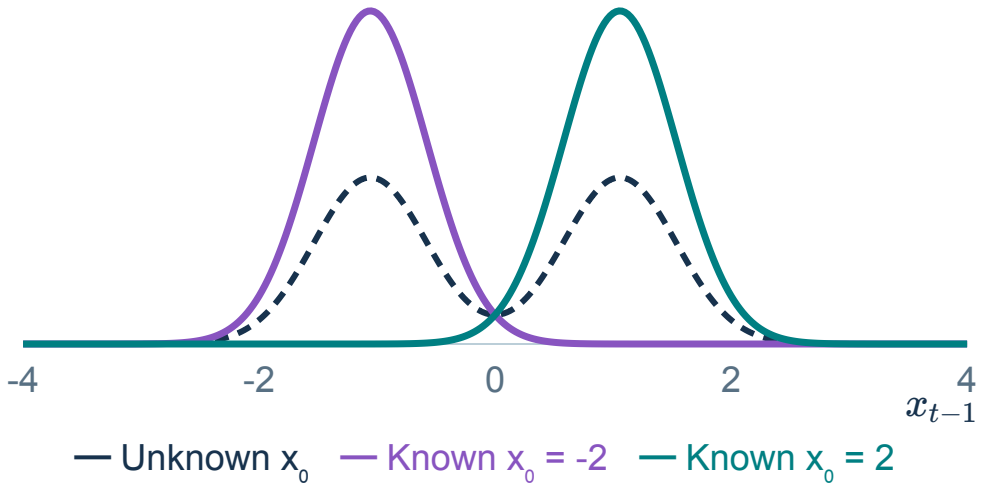
Posterior weight 50.0%

$x_0 = 2$

Posterior weight 50.0%

Unknown x_0: posterior-weighted mixture.
Known x_0: the corresponding Gaussian.

$q(x_{t-1}|x_t)$ and $q(x_{t-1}|x_t, x_0)$



$$q(x_{t-1}|x_t) = \sum_{x_0 \in \{-2,2\}} q(x_{t-1}|x_t, x_0)\, q(x_0|x_t)$$

Exact toy example. During training x_0 is available; during generation it is unknown.

Distribution Summary

Forward process: with a suitable noise schedule, $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$ as the residual signal becomes negligible:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I});$$
$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1-\bar{\alpha}_t) \cdot \mathbf{I}).$$

Reverse process refers to an intractable distribution that can be approximated by a normal distribution (with unknown parameters) for small β_t:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

Conditioned reverse process is a normal distribution with known parameters, describing how to denoise a noisy image $\mathbf{x}_t$ when we know the clean image $\mathbf{x}_0$.

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \cdot \mathbf{I})$$

Noise Schedule through Signal-to-Noise Ratio

Signal and Noise

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}.$$

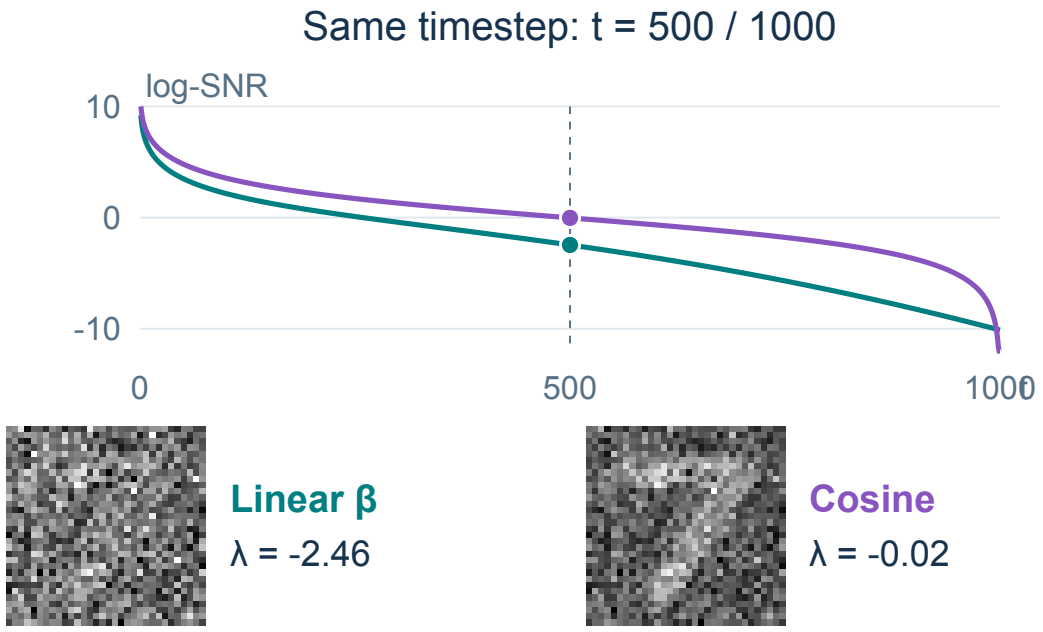
For $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$, compare squared scales:

$$\mathrm{SNR}(t) = \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}.$$

A Single Noise-Level Coordinate

$$\lambda_t = \log \mathrm{SNR}(t), \qquad \bar{\alpha}_t = \frac{1}{1+e^{-\lambda_t}}.$$

$\lambda_t > 0$: signal dominates.
$\lambda_t = 0$: equal signal and noise scales.
$\lambda_t < 0$: noise dominates.



Same clean image and Gaussian noise.
Fixed display scale; log-SNR axis clipped to [−12, 10].

Same timestep, different noise level.

Higher pixel-space resolution: more noise can help; shifting log-SNR down increases corruption.

Kingma D. et al. Variational Diffusion Models, 2021
Nichol A., Dhariwal P. Improved Denoising Diffusion Probabilistic Models, 2021
Hoogeboom E. et al. Simple diffusion: End-to-end diffusion for high resolution images, 2023

Outline

01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

05 Gaussian Diffusion Reparametrization

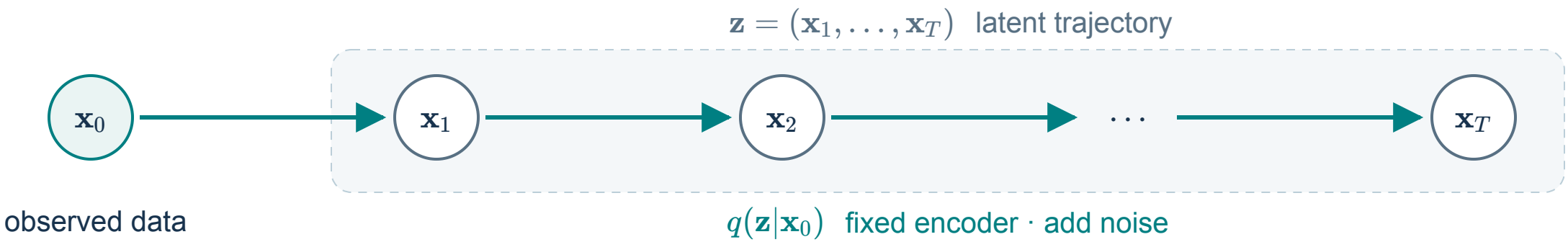
06 Denoising Diffusion Probabilistic Model (DDPM)

Gaussian Diffusion Model as VAE

Let's treat $\mathbf{z} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ as a latent variable (**note:** each $\mathbf{x}_t$ has the same dimension), and $\mathbf{x} = \mathbf{x}_0$ as the observed variable.

Latent Variable Model

$$p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p_{\boldsymbol{\theta}}(\mathbf{z})$$



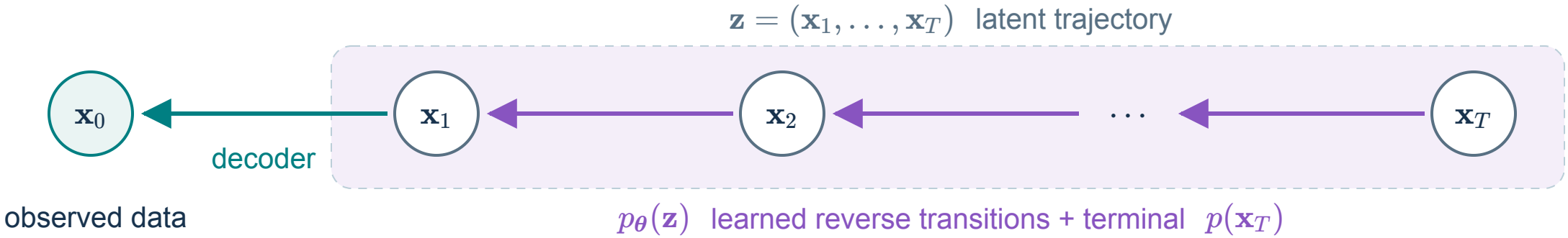
Forward Diffusion

Variational posterior (encoder), with **no learnable parameters**:

$$q(\mathbf{z}|\mathbf{x}) = q(\mathbf{x}_1, \dots, \mathbf{x}_T|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}).$$

Gaussian Diffusion Model as VAE

$$p_{\theta}(\mathbf{x}, \mathbf{z}) = p_{\theta}(\mathbf{x}|\mathbf{z})p_{\theta}(\mathbf{z})$$



Reverse Diffusion

Generative distribution (decoder)

$$p_{\theta}(\mathbf{x}|\mathbf{z}) = p_{\theta}(\mathbf{x}_0|\mathbf{x}_1).$$

Prior distribution

$$p_{\theta}(\mathbf{z}) = p_{\theta}(\mathbf{x}_1, \ldots, \mathbf{x}_T) = \prod_{t=2}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \cdot p(\mathbf{x}_T).$$

Unlike the vanilla VAE: a **simple decoder** $p_{\theta}(\mathbf{x}|\mathbf{z})$ and a **Markov-chain prior** $p_{\theta}(\mathbf{z})$.

Outline

01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

05 Gaussian Diffusion Reparametrization

06 Denoising Diffusion Probabilistic Model (DDPM)

ELBO for Gaussian Diffusion Model

Standard ELBO

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) \geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z}|\mathbf{x})} = \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \to \max_{\boldsymbol{\phi},\boldsymbol{\theta}}$$

Derivation

$$\begin{aligned}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) &= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_0, \mathbf{x}_{1:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \\ &= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T) \prod_{t=1}^{T} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})}\end{aligned}$$

- Let's try to decompose the ELBO into individual KL divergence terms.
- We need to replace $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ with $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ in the denominator.
- Let's condition on $\mathbf{x}_0$ to make the reverse $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ tractable.

ELBO for Gaussian Diffusion Model

Bayes' Rule ($t \geq 2$)

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) = \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}$$

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T)\prod_{t=1}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)\prod_{t=2}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_1|\mathbf{x}_0)\prod_{t=2}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)\prod_{t=2}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_1|\mathbf{x}_0)\prod_{t=2}^{T} \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p(\mathbf{x}_T)p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)\prod_{t=2}^{T} p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_T|\mathbf{x}_0)\prod_{t=2}^{T} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}$$

Markov Property

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) = q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

Conditioning on $\mathbf{x}_0$ leaves the forward transitions unchanged.

Separate the decoder step $t = 1$.

Apply Bayes' rule for each $t \geq 2$.

Intermediate marginals telescope; the $q(\mathbf{x}_1|\mathbf{x}_0)$ factor cancels too.

ELBO for Gaussian Diffusion Model

Derivation (continued)

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\Bigg[\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) + \log\frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^{T}\log\left(\frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}\right)\Bigg]$$

Expand the logarithm: products become sums.

$$= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) + \mathbb{E}_{q(\mathbf{x}_T|\mathbf{x}_0)}\log\frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^{T}\mathbb{E}_{q(\mathbf{x}_{t-1},\mathbf{x}_t|\mathbf{x}_0)}\log\left(\frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}\right)$$

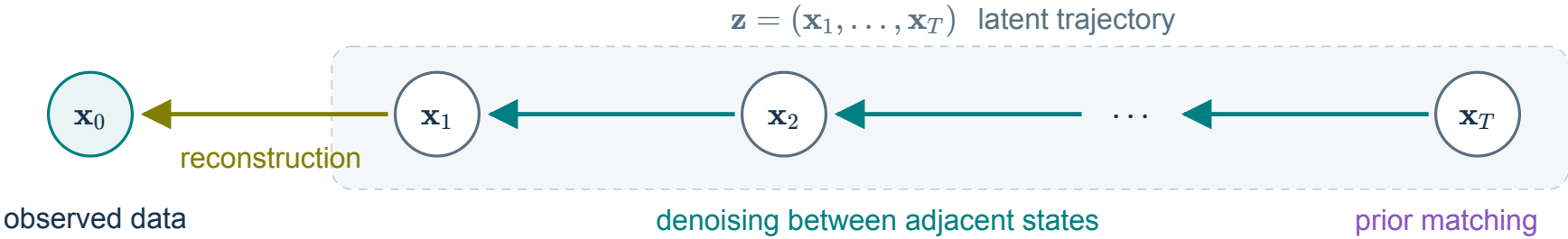
Integrate out unused variables. Each term depends on one state or one adjacent pair.

$$= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T}\underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$

Use the definition of KL. For each $\mathbf{x}_t$, average over $q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)$; then average over $q(\mathbf{x}_t|\mathbf{x}_0)$.

ELBO for Gaussian Diffusion Model

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$



- First term is the decoder distribution, with $\mathbf{x}_1 \sim q(\mathbf{x}_1|\mathbf{x}_0)$:

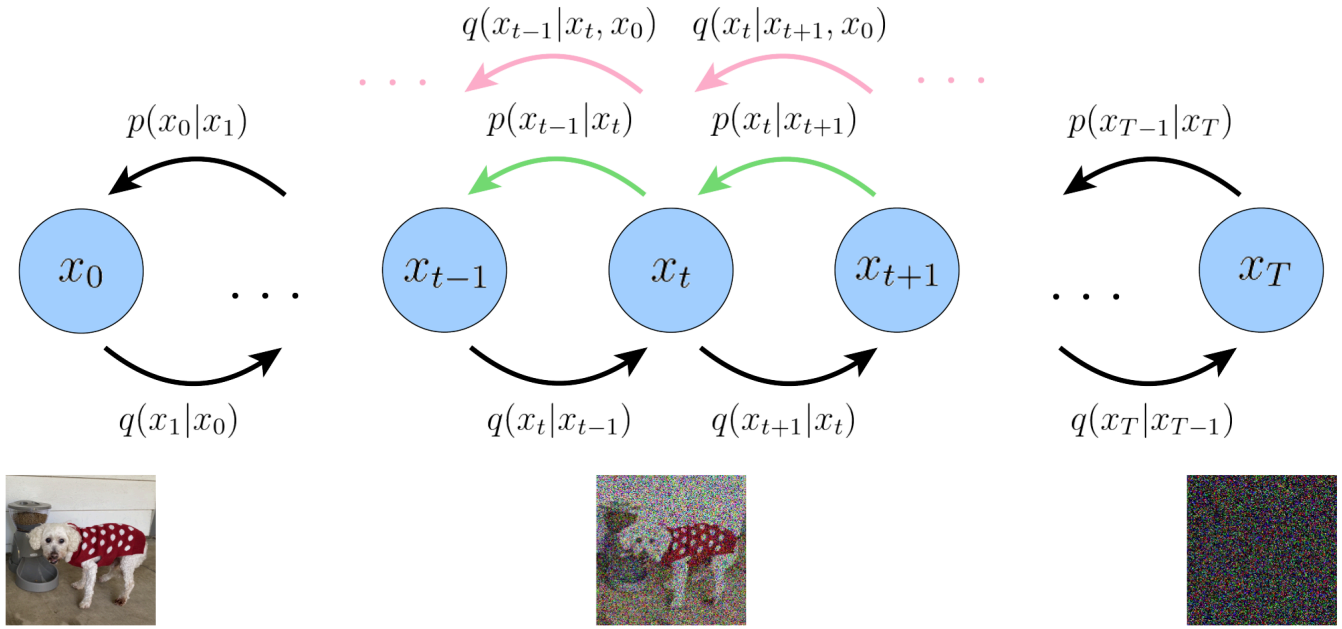
$$\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) = \log \mathcal{N}\big(\mathbf{x}_0|\boldsymbol{\mu}_{\boldsymbol{\theta},1}(\mathbf{x}_1), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},1}(\mathbf{x}_1)\big).$$

- Second term is constant:

$$p(\mathbf{x}_T) = \mathcal{N}(0, \mathbf{I}), \qquad q(\mathbf{x}_T|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_T} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_T) \cdot \mathbf{I}).$$

- Third term is the main contributor to the ELBO.

ELBO for Gaussian Diffusion Model



$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\left(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\right)$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}),$$

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1}|\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

ELBO for Gaussian Diffusion Model

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}),$$

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\big(\mathbf{x}_{t-1}|\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\big)$$

Let's assume that

$$\boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \tilde{\beta}_t \mathbf{I}$$

$$\Rightarrow \quad p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\big(\mathbf{x}_{t-1}|\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}\big).$$

Theoretically, the optimal $\boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)$ lies in $[\tilde{\beta}_t, \beta_t]$:

- β_t is optimal for $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$;
- $\tilde{\beta}_t$ is optimal for $\mathbf{x}_0 \sim \delta(\mathbf{x}_0 - \mathbf{x}^*)$.

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(\mathcal{N}\big(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}\big) \| \mathcal{N}\big(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}\big)\big)$$

$$= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2\right]$$

ELBO for Gaussian Diffusion Model

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$.
3. Compute the ELBO:

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}\left(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)\right) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]}_{\mathcal{L}_t}.$$

Sampling

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

Outline

01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

05 Gaussian Diffusion Reparametrization

06 Denoising Diffusion Probabilistic Model (DDPM)

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[\frac{1}{2\tilde{\beta}_t}\left\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right\|^2\right]$$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\cdot\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\cdot\mathbf{x}_0$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\cdot\boldsymbol{\epsilon} \quad\Rightarrow\quad \mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\cdot\boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}}$$

- There is a linear relationship between $\boldsymbol{\epsilon}$, $\mathbf{x}_t$, and $\mathbf{x}_0$.
- Let's try to rewrite this mean using only $\mathbf{x}_t$ and $\boldsymbol{\epsilon}$.

$$\begin{aligned}\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \boldsymbol{\epsilon}) &= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\cdot\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\cdot\left(\frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\cdot\boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}}\right) \\ &= \frac{1}{\sqrt{\alpha_t}}\cdot\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\cdot\boldsymbol{\epsilon}\end{aligned}$$

Ho J. Denoising Diffusion Probabilistic Models, 2020

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[\frac{1}{2\tilde{\beta}_t}\left\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right\|^2\right]$$

Reparametrization

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}$$

$$\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)$$

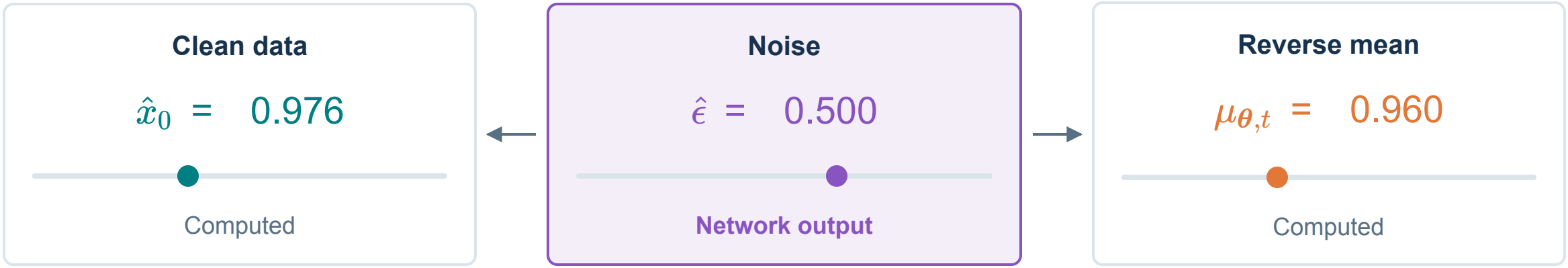
$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[\frac{(1-\alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1-\bar{\alpha}_t)}\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right\|^2\right] \\ &= \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[\frac{(1-\alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1-\bar{\alpha}_t)}\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}\right)\right\|^2\right]\end{aligned}$$

At every step of the reverse process, we attempt to predict the noise $\boldsymbol{\epsilon}$ that was used in the forward diffusion process!

One Prediction, Three Parameterizations

One coordinate. Keep the input and timestep fixed: $x_t = 1, \quad \alpha_t = 0.6, \quad \bar{\alpha}_t = 0.39$

Choose one network output; compute the other two. Example: predict noise.



$$x_t = \sqrt{\bar{\alpha}_t}\hat{x}_0 + \sqrt{1-\bar{\alpha}_t}\hat{\epsilon}$$

$$\mu_{\theta,t} = \frac{x_t}{\sqrt{\alpha_t}} - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\hat{\epsilon}$$

Switch the network output: the values stay the same; only the parameterization changes.

All three are predicted quantities, linked by known conversions.

Diffusion Parameterizations

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}.$$

Network output at $\mathbf{x}_t$	Regression target	Clean-data estimate
Noise $\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}$	$\boldsymbol{\epsilon}$	$\dfrac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}}{\sqrt{\bar{\alpha}_t}}$
Clean data $\hat{\mathbf{x}}_{\boldsymbol{\theta},t}$	$\mathbf{x}_0$	$\hat{\mathbf{x}}_{\boldsymbol{\theta},t}$
Score $\mathbf{s}_{\boldsymbol{\theta},t}$	$-\dfrac{\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}_t}}$	$\dfrac{\mathbf{x}_t + (1-\bar{\alpha}_t)\,\mathbf{s}_{\boldsymbol{\theta},t}}{\sqrt{\bar{\alpha}_t}}$
v-prediction $\mathbf{v}_{\boldsymbol{\theta},t}$	$\sqrt{\bar{\alpha}_t}\,\boldsymbol{\epsilon} - \sqrt{1-\bar{\alpha}_t}\,\mathbf{x}_0$	$\sqrt{\bar{\alpha}_t}\,\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\mathbf{v}_{\boldsymbol{\theta},t}$

Convertible predictions; different losses. Plain MSE assigns different weights to different noise levels.

Salimans T., Ho J. Progressive Distillation for Fast Sampling of Diffusion Models, 2022

Reparametrization of DDPM

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$

$$\mathcal{L}_t = \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[\frac{(1-\alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1-\bar{\alpha}_t)}\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}\big(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}\big)\right\|^2\right]$$

Dropping the time-dependent coefficient **changes the relative weights of noise levels**. The reconstruction term ($t = 1$) also uses noise prediction.

Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}\big(\sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\cdot\boldsymbol{\epsilon}\big)\right\|^2$$

Outline

01 Forward Gaussian Diffusion Process

02 Reverse Gaussian Diffusion Process

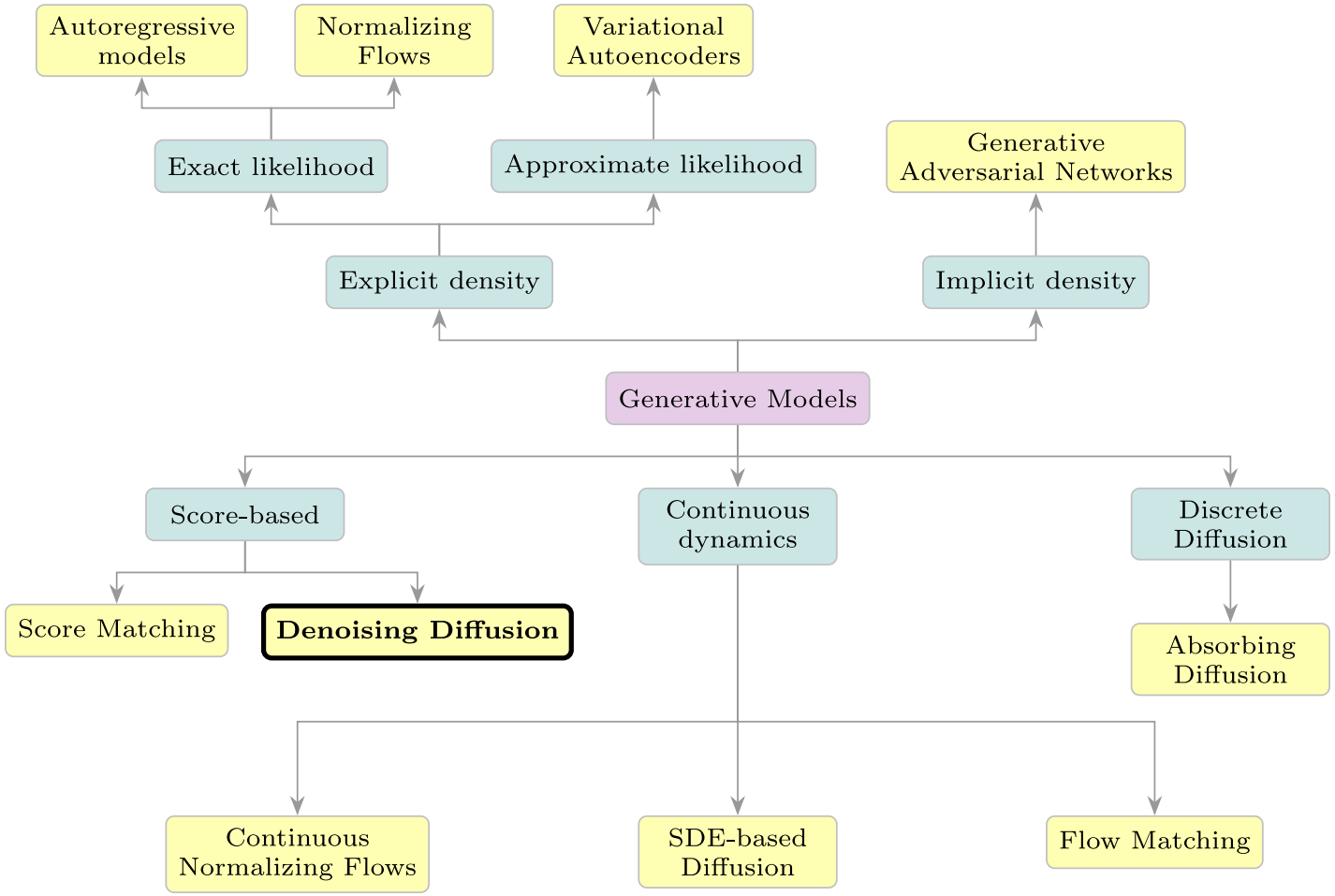
03 Gaussian Diffusion Model as VAE

04 Diffusion ELBO Derivation

05 Gaussian Diffusion Reparametrization

06 Denoising Diffusion Probabilistic Model (DDPM)

Generative Models Taxonomy



Denoising Diffusion Probabilistic Model (DDPM)

DDPM is a VAE Model

- The encoder is a fixed Gaussian Markov chain $q(\mathbf{x}_1, \dots, \mathbf{x}_T | \mathbf{x}_0)$.
- The latent variable is hierarchical (at each step, its dimension equals the input's).
- The decoder is a simple Gaussian model $p_{\boldsymbol{\theta}}(\mathbf{x}_0 | \mathbf{x}_1)$.
- The prior distribution is given by a parametric Gaussian Markov chain $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1} | \mathbf{x}_t)$.

Forward Process

1. $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$;
2. $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}_t$;
3. $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$.

Reverse Process

1. $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$;
2. $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \boldsymbol{\sigma}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\boldsymbol{\epsilon}_t$;
3. $\mathbf{x}_0 \sim p_{\boldsymbol{\theta}}(\mathbf{x}_0) \approx p_{\text{data}}(\mathbf{x}_0)$.

Denoising Diffusion Probabilistic Model (DDPM)

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L}_{\text{simple}} = \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\|^2$.

Sampling (Ancestral)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute the mean of $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$:

$$\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t).$$

3. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

Summary

- Forward diffusion has exact Gaussian conditionals; a suitable schedule removes the signal and controls SNR.
- For small steps, Gaussian reverse transitions approximate the path from noise to data.
- Conditioning on the clean image gives a tractable Gaussian reverse distribution.
- Gaussian diffusion is a VAE with a hierarchy of latent variables and a fixed encoder.
- Its ELBO decomposes into reconstruction, prior matching, and denoising terms; Gaussian denoising terms reduce to squared error.
- The reverse mean can be parameterized through clean data, noise, score or v; plain MSE induces different noise-level weights.
- DDPM reweights the ELBO denoising terms into a simple noise-prediction loss and uses ancestral sampling.